Reattempt A | Version 5 | Solutions

*Each question earns **Pass** or **Not Yet**. Pass (high) and Pass (low) mark the range of passing work; both earn the same Pass.*

Q1 | Opportunity Cost (A1.1)

Suppose Honeydukes, a candy shop in Hogsmeade, can make 10 pounds of Chocolate Frogs *or* 8 pounds of Ice Mice in one day.

> **What to look for:**
>
> **Pass (high):** Both per-unit tradeoffs and the best alternative forgone for each choice.
>
> **Pass (low):** Correct ratios with an arithmetic slip, or an omitted unit made clear by the work.
>
> **Not Yet:** Reversed tradeoffs, adding all alternatives, or missing evidence of the best alternative.

a) What is Honeydukes' opportunity cost of 1 pound of Ice Mice? **5/4 pounds of Chocolate Frogs**

> a) **5/4 pounds of Chocolate Frogs**: giving up 10 pounds of Chocolate Frogs buys 8 pounds of Ice Mice, so one pound of Ice Mice costs $10/8 = 5/4$ pounds of Chocolate Frogs.

b) What is Honeydukes' opportunity cost of 1 pound of Chocolate Frogs? **4/5 pound of Ice Mice**

> b) **4/5 pound of Ice Mice**, the reciprocal.

c) Honeydukes ranks its options for the day from best (1) to worst (3):

1. Sell candy in the shop (S)
2. Cater a Hogwarts feast (C)
3. Restock the cellar (R)

(i) What is Honeydukes' opportunity cost of choosing S? **Catering a Hogwarts feast (C)**

> c) (i) C, its next best option after S.

(ii) What is Honeydukes' opportunity cost of choosing R? **Selling candy in the shop (S)**

> c) (ii) S, its best alternative to R. R, ranked last, is the cost of nothing.

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Q2 | Absolute and Comparative Advantage (A2.1)

Honeydukes and Zonkos are two candy shops in Hogsmeade, the only all-wizarding village in Britain. In one day, Honeydukes can make 10 pounds of Chocolate Frogs or 8 pounds of Ice Mice, and Zonkos can make 20 pounds of Chocolate Frogs or 5 pounds of Ice Mice. Set up a Production Table with both shops' output per day. Then set up an Opportunity Cost table with both shops' opportunity costs for each good. Which shop has the absolute advantage (AA) and comparative advantage (CA) in each good?

What to look for:

Pass (high): Production and cost tables; AA from greater output and CA from lower opportunity cost.

Pass (low): A copying or arithmetic slip and its consistent consequences when the method and comparisons are clear.

Not Yet: Using output to choose CA, systematically reversing costs, or wrong answers with no cost work to show where the error came from. Correct answers earned Pass even without the cost work shown.

Production table: each shop's output per day if it makes only that good, straight from the setup.

	Chocolate Frogs	**Ice Mice**
Honeydukes	10 pounds	8 pounds
Zonkos	20 pounds	5 pounds

Opportunity costs: each cell is what the shop gives up of the other good. Honeydukes: a pound of Chocolate Frogs costs $8/10 = 4/5$ pound of Ice Mice, a pound of Ice Mice costs $10/8 = 5/4$ pounds of Chocolate Frogs. Zonkos: a pound of Chocolate Frogs costs $5/20 = 1/4$ pound of Ice Mice, a pound of Ice Mice costs $20/5 = 4$ pounds of Chocolate Frogs.

	Chocolate Frogs	**Ice Mice**
Honeydukes	4/5 Ice Mice	5/4 Chocolate Frogs
Zonkos	1/4 Ice Mice	4 Chocolate Frogs

AA in Chocolate Frogs: **Zonkos**

AA in Ice Mice: **Honeydukes**

CA in Chocolate Frogs: **Zonkos**

CA in Ice Mice: **Honeydukes**

AA in Chocolate Frogs: **Zonkos** ($20 > 10$). AA in Ice Mice: **Honeydukes** ($8 > 5$).

CA in Chocolate Frogs: **Zonkos** ($1/4 < 4/5$). CA in Ice Mice: **Honeydukes** ($5/4 < 4$). Here each shop holds both advantages in the same good.

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Q3 | Zonkos' PPF (A1.2)

Suppose Zonkos, a candy shop in Hogsmeade, can make 20 pounds of Chocolate Frogs (F) *or* 5 pounds of Ice Mice (M) in one day. There are two graphs, one beside (a) and one beside (b). On both graphs, draw Zonkos' PPF, F on the vertical axis and M on the horizontal. Label the intercepts.

What to look for:

Pass (high): A meaningful PPF, inside/on/outside classifications, and the effect of each capacity change.

Pass (low): An isolated wrong or missing label when the graph and other answers establish the distinction.

Not Yet: No usable PPF graph, repeated category confusion, or a wrong capacity-change answer without supporting evidence.

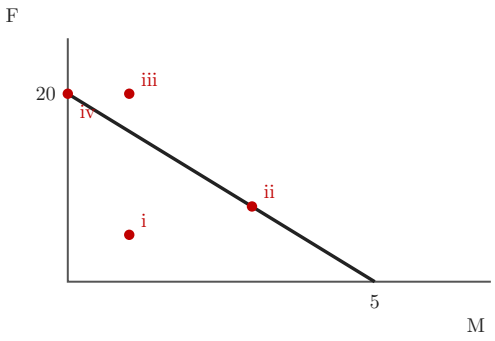
a) For each daily output, write whether it is unattainable, inefficient, or efficient, and mark it on the first graph.

(i) $5F$ and $1M$ **Inefficient**

(ii) $8F$ and $3M$ **Efficient**

(iii) $20F$ and $1M$ **Unattainable**

(iv) $20F$ **Efficient**

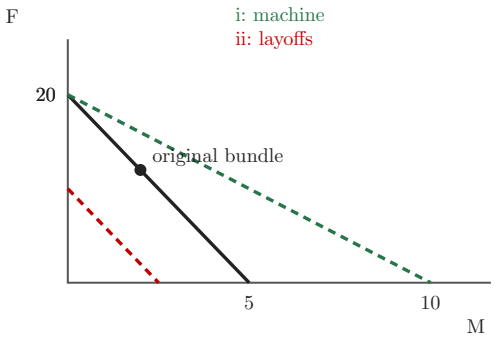


PPF: Straight line from $20F$ to $5M$, so $F = 20 - 4M$. Opportunity cost: $4F$ per M. **a) In order:** Inefficient (at $1M$ the frontier allows $16F$); Efficient ($20 - 4 \cdot 3 = 8$); Unattainable (above the frontier); Efficient (the vertical intercept).

b) Zonkos currently makes $12F$ and $2M$. On the second graph, draw its PPF and mark this output. Is that output unattainable, inefficient, or efficient:

(i) if it buys a more capable Ice Mice machine, doubling its daily output of Ice Mice? **Inefficient**

(ii) if instead investors at Gringotts lay off many of Zonkos' employees? **Unattainable**



b) The original bundle is on the original frontier ($20 - 4 \cdot 2 = 12$). (i) Machine: the M intercept doubles to 10, F stays at 20; at $2M$ the new frontier allows $16F$, so the original bundle is **inefficient**. (ii) Layoffs: both intercepts fall and the frontier shifts in; **unattainable**. **Taylor's Note:** In Q3.b, new frontiers illustrate the shifts but they are not required in the answer.

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Q4 | Specialization and Trade (A3.1)

Honeydukes and Zonkos each make Jelly Slugs and Sugar Quills. Their opportunity costs are listed in the following opportunity cost table.

	Jelly Slug	**Sugar Quill**
Honeydukes	4/3 Sugar Quills	3/4 Jelly Slug
Zonkos	5 Sugar Quills	1/5 Jelly Slug

What to look for:

Endpoints: The bounds identify an open interval. At an endpoint one producer is indifferent.

Pass (high): The lower-cost specializer, the full trade interval in common units, and an acceptable rate.

Pass (low): Correct specializer, bounds, and acceptable selection plus one extra outside checkbox.

Not Yet: Wrong specialization or bounds without correct work, only outside rates selected, or multiple extra selections.

a) If the two shops specialize and trade with each other, which one should specialize in Sugar Quills? **Zonkos**

a) **Zonkos**, the shop with the lower opportunity cost of a Sugar Quill. Honeydukes has the lower cost of a Jelly Slug (4/3 versus 5 Sugar Quills) and Zonkos the lower cost of a Sugar Quill (1/5 versus 3/4 Jelly Slug).

b) What is the range of exchange rates that would make both Honeydukes and Zonkos better off?

Between **4/3** and **5** Sugar Quills for 1 Jelly Slug

b) Between **4/3** and **5** Sugar Quills for 1 Jelly Slug: the two shops' opportunity costs of a Jelly Slug, read down the Jelly Slug column. Below the low end the Jelly Slug maker would rather keep its Jelly Slugs; above the high end the Sugar Quill maker would rather make Jelly Slugs itself.

c) Select all the exchange rates which could facilitate a trade between the two shops.

- ☐ 1 Jelly Slug for 1 Sugar Quill
- ☒ 1 Jelly Slug for 2 Sugar Quills
- ☒ 1 Jelly Slug for 3 Sugar Quills
- ☐ 1 Jelly Slug for 6 Sugar Quills

c) **1 Jelly Slug for 2 Sugar Quills** and **1 Jelly Slug for 3 Sugar Quills**. 1 is below Honeydukes' cost of 4/3, so Honeydukes keeps its Jelly Slugs; 6 is above Zonkos' cost of 5, so Zonkos would rather make Jelly Slugs itself.